Worksheet 10 · Linear Inequalities IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Solve linear inequalities • Represent inequalities on a number line • List integer values that satisfy an inequality		
Grade: / 32	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Linear Inequalities*

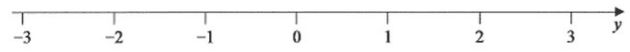
Question 1 (5 marks)

(a) Solve the inequality $4x + 13 \geqslant 27$

.....

(2)

(b) On the number line, represent the inequality $y \geqslant -1$



(1)

n is an integer.

(c) Write down all the values of n that satisfy $-3 < n \leqslant 2$

.....

(2)

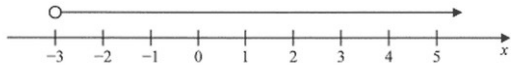
Question 2 (2 marks)

Solve the inequality $5x - 4 < 39$

.....

(2)

Question 3 (3 marks)



Write down the inequality shown on the number line.

.....

(1)

(b) Solve the inequality $4y - 13 \leqslant y + 8$

.....

(2)

Question 4 (3 marks)

Solve the inequality $7 < 4x - 1 \leqslant 17$

(3)

Question 5 (3 marks)

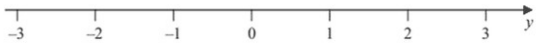
n is an integer.

(a) Write down all the values of n such that $-2 \leqslant n < 3$

.....

(2)

(b) On the number line, represent the inequality $y \leqslant 1$



(1)

Question 6 (4 marks)

(a) On the number line, show the inequality $-2 \leqslant y < 1$



(2)

n is an integer.

(b) Write down all the values of n that satisfy $-3.4 < n \leqslant 2$

.....

(2)

Question 7 (4 marks)

$-4 \leqslant 2y < 6$

y is an integer.

(a) Write down all the possible values of y.

.....

(2)

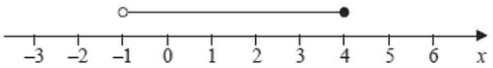
(b) Solve the inequality $7t - 3 \leqslant 2t + 31$

Show your working clearly.

.....

(2)

Question 8 (5 marks)



a) An inequality is shown on the number line. Write down this inequality.

bi) Solve the inequality $2(y - 3) > 1$

ii) Write down the lowest **integer** which satisfies this inequality.

Question 9 (3 marks)

Solve the inequality $3x + 15 < 8x + 3$

Show clear algebraic working.